Local–edge routing for conformational B-cell epitope prediction: what survives leakage-free selection

Anandharajan TRV*
Department of Electronics and Communication Engineering
St. Joseph College of Engineering, Sriperumbudur,
Chennai 602117, Tamil Nadu, India

Abstract

Screening antigens for conformational B-cell epitopes increasingly relies on protein language models. Those models are too heavy for the laboratory workstations where candidate antigens are first triaged, which pushes laboratories towards sending raw sequences to a remote service. We study the intermediate option. Each antigen is routed to local or remote execution according to the local model's own uncertainty. On the same runs we measure predictive quality, latency, modelled carbon and the fraction of antigens whose raw sequence leaves the device, resampling over antigens rather than over re-runs. On the BP3C50ID external split, routing runs at 13.6 ms per antigen against 42.4 ms for always-remote execution, a difference of −28.8 ms (95% CI [-37.9, -19.1]). It transmits the raw sequence of 0.285 of antigens, where always-remote execution transmits all of them. Its effect on predictive quality is not resolved at this sample size. Against always-remote execution all four quality metrics have intervals containing zero, and against always-local execution only F1 separates them (ΔF1 +0.033 [+0.004, +0.075]). The same comparisons appear decisive under the seed-only intervals that repeated benchmarking conventionally reports, which on a fixed 15-antigen split measure re-run dispersion rather than sampling uncertainty. The cost accounting runs the other way: in our model routing is the most carbon-intensive policy, for any positive cost constants, though the routed fraction at which that begins depends on constants we assume rather than measure. We give in closed form the routed fraction above which this must occur, 0.139 here. We then re-examine an earlier unpublished result of our own, that fusing our local predictor with BepiPred-3.0 improves on BepiPred-3.0 alone. Reproducing the original protocol to within 4.9e-07 across all 44 of its stored grid entries, we show that the mixing weight and decision threshold were selected on the evaluation split. Under leave-one-sequence-out nested selection the advantage is no longer detectable on BP3C50ID (ΔROC-AUC -0.0005 [-0.0033, +0.0021]), and the MCC estimate changes sign, from +0.0115 to -0.0329. A second arm on a 47-antigen IEDB subset behaves the same way (+0.0008 [-0.0010, +0.0027]), though it fuses against our own local predictor rather than BepiPred-3.0 and so tests a different pair. In both arms the folds place almost all weight on the reference predictor. These sample sizes cannot establish that fusion does not help; what they do show is how far the selection step moves the estimate. On benchmarks of this size, the conventional analyses report more confidence than the data supports.

Keywords: conformational B-cell epitope prediction; edge inference; model selection; data leakage; reproducible benchmarking

1 Introduction

A laboratory that has just expressed a new antigen and wants to know where its surface-exposed B-cell epitopes are has good computational options, and a practical problem in using them. The best current conformational predictors are built on protein language models whose weights run

*Corresponding author. E-mail: trvanandharajan@gmail.com

to billions of parameters (Lin et al., 2023). Running one well needs either a workstation with a capable accelerator or, far more commonly, uploading the sequence to a hosted service. Neither is comfortable for every group, and accelerators are unevenly distributed. For antigens from unpublished constructs, from clinical isolates, or from an industrial pipeline, the sequence itself is the sensitive object: sending it to a third-party endpoint is a disclosure decision rather than a technical one.

The obvious middle path is to keep a small model on the local machine and call the remote service only when the local model is unsure. Systems research has established the idea under names such as split computing, model cascades and early exiting (Matsubara et al., 2022). Its appeal in a screening setting is easy to state: most antigens should be easy, the easy ones never leave the device, and the hard minority gets the benefit of the larger model. What such a scheme costs and returns is much less clear. A laboratory cares about three quantities at once: how good the predictions are, how long screening takes, and how much sequence leaves the building. We measure all three on the same runs rather than arguing them separately.

We build a measurement harness that runs a set of routing policies, from always-local through uncertainty-triggered routing to always-remote, on identical inputs with identical seeds. It records predictive quality, latency, data movement and an energy model per antigen. Over 10 seeds on the BP3C50ID external split, the answer arrives with intervals attached: routing on local uncertainty gives up nothing detectable in quality against always-remote execution while cutting latency to roughly a third and keeping most raw sequences on the device. It also gives an answer nobody in this project expected. A routed antigen is charged twice: once for the local pass that decides it is hard, once for the remote pass that then processes it. Routing is therefore the most carbon-intensive policy in the comparison. The green framing that usually accompanies edge inference does not survive contact with the accounting, and Section 4.1 gives the condition under which it would.

The paper also re-examines an earlier unpublished analysis of our own, which reported that blending the scores of our local predictor with those of BepiPred-3.0 (Clifford et al., 2022) improved on BepiPred-3.0 alone. The mixing weight and the decision threshold had both been chosen by scanning the evaluation split and keeping the configuration that scored best. Kapoor and Narayanan (2023) identify that step as the most common form of leakage in applied machine learning. It is easy to commit, because a sensitivity analysis and a model-selection procedure look superficially alike. We reproduce the original numbers under the original protocol, to within 4.9e-07. We then re-estimate the same quantity with weight and threshold chosen inside the training folds of a leave-one-sequence-out loop. The advantage does not survive on either dataset. On BP3C50ID the difference against BepiPred-3.0 is no longer detectable and the MCC estimate changes sign; on a larger IEDB-derived subset, which fuses against our own local predictor rather than BepiPred-3.0, the difference is likewise indistinguishable from zero. On both, the selection procedure places almost all of the weight on the reference predictor rather than on ours. Neither dataset is large enough to show that fusion cannot help, and we do not claim that it does not.

The literature this belongs to is accumulating fusion and ensemble results faster than it scrutinises how their combination parameters were chosen, and the corrective is cheap: the nested protocol here costs one extra loop over sequences and needs no new data.

Contributions. Concretely, this paper offers the following.

1. A reproducible measurement harness for conformational B-cell epitope screening. It records predictive quality, latency, data movement and modelled energy together, per antigen and per seed, and states the provenance of each cost quantity (Section 3).

2. A 10-seed characterisation of five local–edge routing policies on BP3C50ID, with bootstrap intervals and paired sign tests. It includes a closed-form condition locating where the known offloading energy threshold (Kumar and Lu, 2010) falls for this task: routing increases

modelled carbon once the routed fraction exceeds 0.139, and a quality-tuned policy routes about twice that often (Section 4.1).

3. An instance, in conformational epitope prediction, of the selection-bias mechanism documented in bioinformatics since Varma and Simon (2006): we reproduce a previously reported fusion advantage under its original, test-derived protocol, then re-estimate it leakage-free on two datasets, where it is no longer detectable and the estimate changes sign (Section 4.3).

4. The observation that, given a free choice on either dataset, the selection procedure assigns our predictor almost no weight. We report the per-fold selections rather than summarising them away (Section 4.4).

2 Background

2.1 Conformational B-cell epitope prediction

A B-cell epitope is the part of an antigen that an antibody binds. Linear epitopes are contiguous in sequence; conformational epitopes, which are the large majority of what antibodies actually recognise, are assembled from residues that the folded structure brings together but that may be far apart in the chain. Prediction is therefore a per-residue binary labelling problem with two awkward properties: positives are a minority of residues, and the label depends on structural context that the sequence encodes only implicitly.

Methods have moved through three broad generations. Propensity-scale approaches scored residues with physicochemical heuristics. Structure-based methods used solved or modelled structures to compute accessibility and protrusion. The current generation feeds representations from a pre-trained protein language model into a supervised head, which is how BepiPred-3.0 (Clifford et al., 2022) and structure-aware successors such as DiscoTope-3.0 (Høie et al., 2024) are built. Recent additions include the SEMA 2.0 web platform (Ivanisenko et al., 2024) and residue-level predictors built directly on ESM-2 projections (Israeli and Louzoun, 2024). Both sit close to the PLM family evaluated here. Gabellieri et al. (2026) survey the field as it now stands. We use BepiPred-3.0 as the reference predictor on BP3C50ID, where it is the strongest openly runnable sequence-based baseline and the method a laboratory would realistically be choosing between. On the IEDB-derived subset we do not have BepiPred-3.0 predictions, so the reference there is our own local-only predictor. The two fusion arms therefore test different pairs, and we keep them separate throughout.

Evaluation in this area is dominated by the BP3C50ID benchmark distributed with BepiPred-3.0, whose external test split holds 15 antigens (2670 residues, 507 of them positive). That is a small split, and its size is worth keeping in view when reading any comparison built on it, including ours. We report the Matthews correlation coefficient alongside ROC-AUC and average precision, because MCC behaves sensibly under the class imbalance that per-residue epitope labelling produces (Chicco and Jurman, 2020). Under that imbalance, accuracy and F1 can both be high for a predictor that has learned little.

2.2 Splitting inference between a local device and a remote service

Running part of a model locally and part remotely is an old idea with a current name. The systems literature treats it as a scheduling problem. The task is to decide where each input is processed, given a local model, a remote model, and a per-input signal about difficulty, so that some combination of accuracy, latency, cost and bandwidth is optimised (Matsubara et al., 2022). The signal is usually the local model's own confidence. That makes the mechanism attractively simple, and it also introduces its characteristic failure: the decision is only as good as the calibration of the model making it.

Whether offloading saves energy at all is a settled question in that literature. The answer depends on the ratio of computation offloaded to the overhead of offloading it. Below a threshold ratio, offloading costs more energy than it saves (Kumar and Lu, 2010). The same effect has been observed recently in model cascades, where a draft-and-verify decoding scheme consumed more energy than running the large model alone despite reducing latency (Dutta et al., 2026). Our contribution on this axis is not the phenomenon but the location of the threshold for this task, and the observation that per-residue epitope screening sits on the unfavourable side of it.

Two things are different in the setting we study. The first is that the quantity a laboratory wants to minimise is often not bandwidth or cost but disclosure, meaning how many of its sequences are transmitted at all. That is a counting measure rather than a continuous one, and no accuracy-oriented objective captures it. The second is that the per-antigen compute is small enough that the routing decision is not free relative to the work it schedules. In large-model serving, a confidence check is negligible beside the inference it gates. In per-residue epitope screening on short sequences, it is not. The carbon results in Section 4.1 are a consequence of that ratio.

2.3 Accounting for energy and carbon

Reported carbon figures for machine-learning systems are rarely measurements. The common practice, which we follow and label as such, is to derive energy from a device power figure and a measured runtime. That energy is then converted to mass of carbon dioxide equivalent with a grid-intensity constant (Strubell et al., 2019; Lacoste et al., 2019; Patterson et al., 2021). Our energy model follows the formulation of Green Algorithms (Lannelongue et al., 2021), which also supplies the amortised-hardware term. That formulation has been applied to bioinformatics workloads specifically (Grealey et al., 2022). We claim no novelty in the instrument. The result is a model whose output is proportional to measured latency by construction. It is useful for comparing policies that run on the same hardware, since the proportionality constant cancels; it is not a measurement of what any particular laboratory's equipment would draw, and we are careful not to present it as one. Table 4 records which of our reported quantities are measured and which are modelled.

2.4 Selection on the evaluation split

When a method has a free parameter — an ensemble weight, a decision threshold, a fusion coefficient — and that parameter is chosen by trying candidates and keeping whichever scores best on the data the paper reports, the reported score is biased upward. The bias grows with the number of candidates tried and shrinks with the size of the evaluation set. This is not a new observation. Varma and Simon (2006) demonstrated it for bioinformatics classifiers two decades ago and showed that nested resampling removes the bias. Boulesteix (2009) named the resulting pattern over-optimism. Jelizarow et al. (2010) showed that re-evaluating a published positive result without its optimistic selection step can remove the advantage entirely. Kapoor and Narayanan (2023) survey how widely the problem persists across applied machine-learning research and how often it is invisible in the write-up, because a table of candidate values reads as a sensitivity analysis whether or not the reported configuration was picked from it. The distinction that matters is not whether the sweep is shown but whether the reported configuration was chosen using the labels it is then scored against. Section 3.6 sets out the two protocols we compare precisely enough for the difference to be checked.

3 Materials and methods

3.1 Datasets

The primary benchmark is BP3C50ID, the split distributed with BepiPred-3.0, in which training and external test antigens are separated at 50% sequence identity. The external test split contains

15 antigens, 2670 residues and 507 positive residues, a positive rate just under one residue in five. All BP3C50ID results below are on this external split; no model or policy parameter is fitted on it.

A second evaluation uses a subset derived from the Immune Epitope Database (Vita et al., 2024), holding 47 test antigens (14408 residues, 2169 positive) alongside 205 training antigens. It is drawn from the `IEDBSeqsNotSharedAt20ID.fasta` collection distributed alongside BP3C50ID, which holds 3560 IEDB sequences selected not to share more than 20% identity with the BP3 data, and is then restricted to sequences of 45–595 residues so that the full policy grid remains runnable on a workstation. Every sequence in the subset is present verbatim in that source collection.

Its purpose is to test whether conclusions drawn from 15 antigens survive on a different and larger collection, which requires the two to be genuinely independent. Independence holds by construction through the identity filter. We also tested it directly against all 358 BP3C50ID sequences and found 0 shared identifiers, 0 shared sequences, and a maximum 10-mer containment of 0.000. That containment means no antigen in the subset shares even a single 10-residue run with any BP3C50ID antigen.

3.2 Predictors

Two families of local and remote predictor are evaluated, and the distinction between them is held fixed across every policy.

The *heuristic family* uses handcrafted per-residue sequence-context features. Its local member is a lightweight residue classifier; its remote member is an extremely randomised trees refiner that also blends the local score and applies sequence smoothing, with the blend weight and smoothing parameters chosen on a validation split.

The *PLM family* replaces the features with frozen embeddings from ESM-2 (Lin et al., 2023), specifically the `esm2_t6_8M_UR50D` checkpoint. The local member places a logistic residue head on those embeddings; the remote member adds contact-derived features and the same trees refiner. A region-constrained variant post-processes the remote probabilities into contiguous regions using contact density, with the region parameters selected on validation data. We use the 8M-parameter checkpoint because it is what runs on the hardware available for this study. It is the smallest of the released ESM-2 models. The PLM results here should therefore be read as a lower bound on what the family can do, not as a characterisation of language-model-based epitope prediction in general.

3.3 Routing policies

A routing policy decides, per antigen, whether the remote model is invoked. Five are compared in each family.

local-only The local model answers every antigen. No sequence is transmitted.

edge-only Every antigen is sent to the remote model.

adaptive-split The local model runs first; if its mean per-residue uncertainty exceeds a threshold, the antigen is re-processed remotely and the remote prediction is used. The threshold and a routing budget are selected on the validation split from a grid of 91 candidate values, subject to a maximum routed fraction of 0.50.

carbon-guard As adaptive-split, with an additional constraint that suppresses routing when the modelled carbon cost of the remote call is not justified by the expected quality gain.

conformal-guard (PLM family) As adaptive-split, with a conformal criterion (Angelopoulos and Bates, 2021) replacing the raw uncertainty threshold.

Every selectable parameter in this section is chosen on a validation split carved out of training data at a fraction of 0.20, grouped by sequence so that no antigen contributes residues to both sides. No routing parameter sees the external test split.

3.4 Cost accounting

Four cost quantities are recorded per antigen, and they differ in how directly they are observed; Table 4 states this for each.

Latency is measured, as wall-clock model-prediction time on the executing device. Bytes transferred are computed from the raw sequence length. Raw-sequence share rate is a count: the fraction of test antigens whose sequence is transmitted at all, which is the quantity a laboratory worried about disclosure would actually audit.

Energy and carbon are modelled, not metered. Operational energy is the product of a fixed device power figure and the measured latency; operational carbon converts that with a grid-intensity constant of $708\,\mathrm{g\,kW^{-1}\,h}$; embodied carbon is a per-request amortisation of a device lifetime figure. The local device is modelled at 34 W and the remote at 145 W.

The embodied term also carries a device-utilisation fraction, applied so that a request is charged only the share of the device's lifetime footprint it actually occupies: 0.20 locally and 0.65 remotely. These reduce the per-request embodied charge from 0.6000 to 0.1200 gCO_2e locally and from 0.2125 to 0.1381 remotely. We state them explicitly because they are the single largest determinant of every carbon figure in this paper. As Section 6 discusses, the comparison between policies is not fully insensitive to them.

The balance between those two terms matters for reading the results, so we state it plainly. At the latencies observed here the operational term is a very small part of each total: a local pass contributes about 7.0e-06 gCO_2e and a remote pass about 2.4e-03, against per-request totals near 0.120 and 0.139. The modelled carbon of a policy is therefore governed almost entirely by how many device-requests it charges, through the amortised embodied term, and only marginally by how long those requests take. Carbon differences between policies here are differences in request counts, not in compute. We make no claim about metered power anywhere in this paper.

3.5 Evaluation protocol for routing policies

The full pipeline (validation split, parameter selection, training and external-split evaluation) is repeated for 10 seeds (42–51).

Repeating a pipeline over seeds on a fixed test split answers a narrow question: how much a number moves when the same antigens are scored again. It says nothing about how the number would move on different antigens, which is what a reader wants to know. Our primary analysis therefore resamples both. For each of 2000 replicates we draw 15 antigens with replacement and 10 seeds with replacement, recompute every policy's pooled residue-level metrics on that resampled set, and take the 2.5th and 97.5th percentiles of the resulting distribution as a 95% interval. One antigen draw is shared across all policies within a replicate, so paired differences stay paired. Intervals are percentile intervals, with no bias correction or studentisation, and the resampling is driven by a fixed seed recorded in the released artifact.

We report seed-only intervals alongside these in Section 4 only to show how different the two are; no claim in this paper rests on them, and the exact sign test over seeds that an earlier version of this work relied on is not used.

Two cautions apply to how we read these intervals. First, with 15 antigens a percentile cluster bootstrap is known to under-cover for non-linear statistics such as MCC and ROC-AUC, so the intervals should be read as indicative rather than exact. Second, several of the quantities we report are not random in sign. A policy that routes only some antigens cannot be slower than one that routes all of them, and cannot transmit more sequences, so for latency and disclosure the interval describes the magnitude of a difference whose direction is fixed by construction. Reporting that such an interval excludes zero would assert nothing. We say so where it applies, and we reserve

the language of detection for the quality metrics, where the sign is genuinely in question. There we treat an interval containing zero as no detected difference, and say so rather than describing such a comparison as a trend.

3.6 Two protocols for selecting a fusion weight

Score fusion combines the reference predictor's per-residue score with ours (CARBON-BCE-R, the local-only predictor) as a convex blend governed by a weight w. The two predictors are calibrated differently, so a blend of their raw outputs would be dominated by whichever has the wider spread. We therefore map each onto $[0, 1]$ through its own empirical distribution function before blending. That distribution is estimated on the training folds and applied to the held-out fold, like any other fitted transform; it is not a rank transform computed over the whole split, which would let held-out residues influence their own normalisation. One consequence is worth stating, because it is easy to assume otherwise. A transform fitted on training data is not exactly rank-preserving on held-out data, so a fold's ROC-AUC after mapping can differ slightly from the same predictor's raw ROC-AUC. The effect is small, under 10^{-3} per fold in our runs, but it is not zero. For this reason we report per-fold paired differences rather than pooled ranking metrics. A decision threshold then converts blended scores into calls. Both w and the threshold are free parameters. The whole question is where they come from.

Test-derived selection. The protocol used in the earlier version of this work evaluates a grid of weights on the evaluation split, reports the row that scores best, and sets the threshold so the blended predictor emits the same number of positive calls that the reference predictor emits *on that same split*. Both quantities are thus functions of the test labels.

Nested selection. Antigens are the unit of resampling. For each held-out antigen in turn, the weight is chosen from the same grid and the threshold is fitted as a value on the training folds' blended scores; both are then applied, unchanged, to the held-out antigen. The operating point is the positive-call rate the reference predictor produces at its own published threshold, recomputed on each training split. Nothing derived from the held-out antigen enters the configuration applied to it: not its labels, not its score distribution, not its own quantiles. That is what makes the resulting predictor deployable, since it needs no knowledge of a new antigen before scoring its first residue. This yields one out-of-fold prediction per antigen with no test label having influenced the configuration applied to it. We report the mean per-antigen difference against the reference predictor with a 95% bootstrap interval over antigens, a Wilcoxon signed-rank test and an exact sign test, and we report the weight chosen in every fold rather than only its median.

Both protocols are implemented in the same module against the same loaders and metric code, so that a difference between their outputs is attributable to the protocol rather than to an implementation difference. Section 4.2 verifies this by reproducing the earlier figures under the test-derived protocol, to within 4.9e-07.

4 Results

4.1 What routing buys, and what it costs

Table 1 reports every policy on the BP3C50ID external split across 10 seeds, and Table 2 gives the paired comparisons that the claims below rest on.

Those five comparisons were fixed before the resampling was run: each contrasts a routing policy with the always-local or always-remote baseline it is meant to improve on, within its own predictor family. The resampling itself produces intervals for all 36 orderable pairs of policies on each metric, or 144 intervals on the four quality metrics together. We mark intervals that exclude zero in Table 2 as a reading aid for those five rows, and no multiplicity correction is applied. A

Table 1: Routing-policy performance on the BP3C50ID external split under antigen-level resampling. Intervals are 95% two-way bootstrap intervals over antigens and seeds (15 antigens, 10 seeds, 2000 resamples), so they carry the uncertainty that matters for a claim about other antigens. Carbon is a model-based estimate, not a metered measurement (Section 3.4).

Policy	MCC	ROC-AUC	Latency (ms)	Raw-sequence share
local-only	0.167 [0.100, 0.229]	0.659 [0.596, 0.712]	0.4 [0.4, 0.5]	0.000 [0.000, 0.000]
edge-only	0.209 [0.150, 0.266]	0.678 [0.621, 0.726]	42.4 [37.2, 48.2]	1.000 [1.000, 1.000]
adaptive-split	0.192 [0.121, 0.255]	0.673 [0.611, 0.724]	13.6 [4.4, 24.0]	0.285 [0.087, 0.507]
carbon-guard	0.165 [0.091, 0.233]	0.658 [0.595, 0.711]	6.6 [1.0, 14.5]	0.133 [0.013, 0.293]
PLM local-only	0.195 [0.124, 0.259]	0.678 [0.614, 0.739]	29.6 [17.2, 51.0]	0.000 [0.000, 0.000]
PLM edge-only	0.209 [0.133, 0.280]	0.691 [0.623, 0.755]	68.3 [54.3, 87.6]	1.000 [1.000, 1.000]
PLM region-constrained	0.219 [0.132, 0.310]	0.690 [0.620, 0.760]	68.3 [54.3, 87.6]	1.000 [1.000, 1.000]
PLM adaptive-split	0.199 [0.124, 0.268]	0.680 [0.615, 0.739]	47.0 [29.3, 70.5]	0.273 [0.107, 0.473]
PLM conformal-guard	0.199 [0.124, 0.268]	0.680 [0.615, 0.739]	47.0 [29.3, 70.5]	0.273 [0.107, 0.473]

Table 2: Paired differences between routing policies under antigen-level resampling, with 95% two-way bootstrap intervals. All four quality metrics are shown for every comparison. An interval containing zero is reported as no detected difference; intervals excluding zero are marked *.

Comparison	ΔMCC	ΔROC-AUC	ΔAP	ΔF1
adaptive-split vs edge-only	-0.017 [-0.066, +0.026]	-0.005 [-0.040, +0.021]	-0.005 [-0.029, +0.027]	+0.002 [-0.030, +0.034]
adaptive-split vs local-only	+0.025 [-0.011, +0.065]	+0.014 [-0.004, +0.038]	+0.015 [-0.001, +0.041]	+0.033* [+0.004, +0.075]
carbon-guard vs adaptive-split	-0.027* [-0.064, -0.001]	-0.015 [-0.038, +0.000]	-0.015* [-0.040, -0.000]	-0.016* [-0.039, -0.001]
PLM region-constrained vs PLM local-only	+0.024 [-0.024, +0.073]	+0.012 [-0.011, +0.033]	+0.018* [+0.003, +0.036]	+0.014 [-0.020, +0.049]
PLM adaptive-split vs PLM edge-only	-0.010 [-0.038, +0.016]	-0.011* [-0.022, -0.001]	-0.014* [-0.026, -0.005]	-0.006 [-0.027, +0.012]

reader should not treat the marks as a family-wise claim: across 144 intervals a handful would fall outside zero under a global null, and several of the marked endpoints sit close enough to zero that the Monte Carlo error of a 2000-resample bootstrap could move them. We draw no conclusion from any single mark, and the conclusions below rest on the size and direction of the differences rather than on which intervals happen to exclude zero.

Two points about those intervals before the results. They are two-way bootstrap intervals over antigens and seeds (2000 resamples), not intervals over seeds alone. The distinction is not cosmetic: the test split holds the same 15 antigens in every seed, so a seed-only interval measures how much a number moves when the harness is re-run and says nothing about how it would move on different antigens. On this data seed-only intervals are roughly an order of magnitude narrower: local-only MCC is 0.167 [0.100, 0.229] under antigen resampling against a seed-only width of a hundredth. For two of the policies the seed-only ROC-AUC interval has width zero, because the underlying predictor is deterministic and only the validation split moves. Every interval below is the antigen-level one. We report it this way because the claims are claims about antigens.

The first result is that the latency and disclosure benefits of routing are large and survive this resampling, while the quality questions do not resolve at this sample size. Figure 1 plots predictive quality against mean per-antigen latency for each policy family.

Against always-remote execution, adaptive-split cuts mean latency by -28.8 ms [-37.9, -19.1] and reduces the fraction of antigens whose raw sequence is transmitted by -0.715 [-0.913, -0.493]

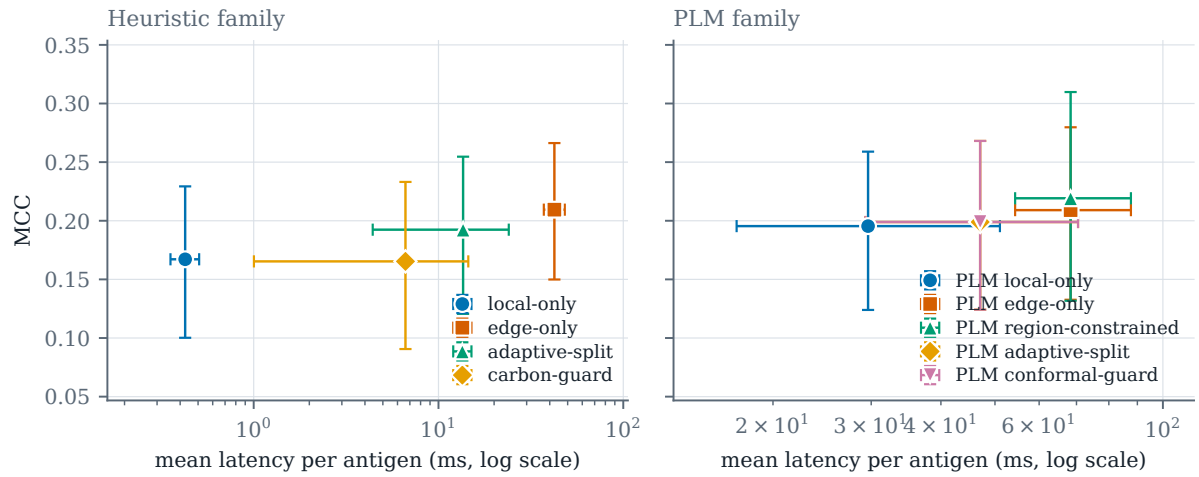


Figure 1: Predictive quality against mean per-antigen latency on the BP3C50ID external split, by policy family. Points are means over 10 seeds; bars are 95% bootstrap intervals on both axes. In the PLM panel the adaptive-split and conformal-guard markers coincide exactly, because the conformal criterion never altered a routing decision on this split.

(Figure 2 shows quality against that fraction). Neither difference could have come out positive: a policy that routes some antigens cannot be slower than one that routes all of them, and cannot transmit more sequences. These intervals therefore say how large the gain is on other antigens, not whether there is one. On quality it differs by ΔMCC -0.017 [-0.066, +0.026], ΔROC-AUC -0.005 [-0.040, +0.021], ΔAP -0.005 [-0.029, +0.027] and ΔF1 +0.002 [-0.030, +0.034]. All four contain zero.

We want to be careful about what that does and does not license. It is not a demonstration of equivalence. With 15 antigens the interval on ΔMCC spans roughly ±0.05, so a quality loss large enough to matter in practice would not be detected here. The defensible statement is that routing's quality cost relative to always-remote execution is not resolved at this sample size, while its latency and disclosure benefits are.

The comparison against always-local execution tells the same story from the other side. ΔMCC is +0.025 [-0.011, +0.065], ΔROC-AUC +0.014 [-0.004, +0.038] and ΔAP +0.015 [-0.001, +0.041], all containing zero, while ΔF1 is +0.033 [+0.004, +0.075], which does not. One metric of four separates them. Under seed-only resampling the same comparison appears decisive on every metric, with ΔMCC +0.034 [+0.022, +0.045] and a sign test over seeds at $p = 0.0039$. That sign test is a statement about ten re-runs on fifteen fixed antigens, and we no longer rest anything on it.

Two comparisons do survive. The carbon-guard policy is worse than adaptive-split on three of four metrics (ΔMCC -0.027 [-0.064, -0.001]), so its latency and carbon savings are bought with quality. Within the PLM family, always-remote execution beats always-local on ROC-AUC and average precision. PLM routing gives part of that back (ΔROC-AUC -0.011 [-0.022, -0.001] against PLM always-remote): in the family where the remote model is genuinely stronger, routing's quality cost is detectable even here.

Table 3 reports the robustness axes the harness provides. The informative row is the one where the remote tier is unavailable: every adaptive-versus-local difference is then exactly zero, because routing degrades to local-only behaviour rather than failing. The grid-intensity axes move nothing that matters, and the parameter the carbon results are most sensitive to — the embodied amortisation constant — is not among the available axes (Section 6).

The second result runs against the expectation that motivated the policy. Figure 3 shows mod-

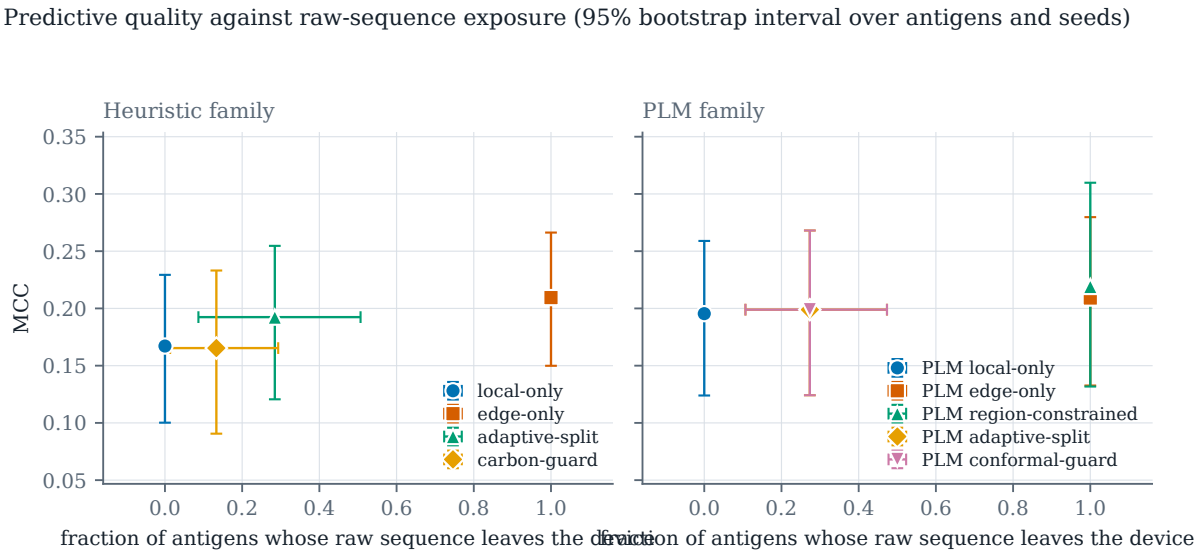


Figure 2: Predictive quality against the fraction of antigens whose raw sequence is transmitted. Policies at zero on the horizontal axis never transmit; edge-only transmits every antigen. Adaptive-split occupies the interior of this trade-off, which always-local and always-remote execution cannot.

Table 3: Sensitivity of the adaptive-split comparisons to the scenario axes available in the harness. Values are mean ΔMCC over seeds. The grid-intensity axes do not touch predictive quality by construction and are included to show which axes were varied; the remote-tier-unavailable row is the informative one, where routing degrades exactly to always-local behaviour. The dominant carbon parameter, the embodied amortisation constant, is not among the available axes.

Scenario	ΔMCC vs local-only	ΔMCC vs edge-only
baseline	+0.034	-0.010
remote tier unavailable	+0.000	+0.000
network +50 ms	+0.034	-0.010
network +5 ms	+0.034	-0.010
grid intensity ×1.5	+0.034	-0.010
grid intensity ×0.5	+0.034	-0.010

elled carbon per antigen for every policy. Adaptive-split is the *highest* of the nine at 0.160 gCO_2e [0.150, 0.167], above both edge-only (0.139) and local-only (0.120). The paired difference against local-only is +0.040 [+0.031, +0.047] and against edge-only +0.021 [+0.010, +0.028], both intervals excluding zero.

This ordering is not an empirical discovery. Because a routed antigen incurs the local pass that decided to route it and then the remote pass, the modelled carbon of the routing policy is determined by the two single-tier costs and the routed fraction r:

$$C_{\text{adaptive}} = C_{\text{local}} + r\,C_{\text{edge}}. \tag{1}$$

Equation 1 reproduces the measured aggregate to within 0.020% (predicted 0.160 against measured 0.160). Rearranging, routing costs more than always-remote execution exactly when

$$r > 1 - \frac{C_{\text{local}}}{C_{\text{edge}}}, \tag{2}$$

which for this device pair is 0.139. The policy routes 0.287 of antigens, roughly twice that threshold, so it lands on the unfavourable side.

Stated this way the result transfers. Equation 2 says routing saves modelled carbon only where the local tier is cheap relative to the remote tier *and* the routed fraction stays below $1-C_{\text{local}}/C_{\text{edge}}$.

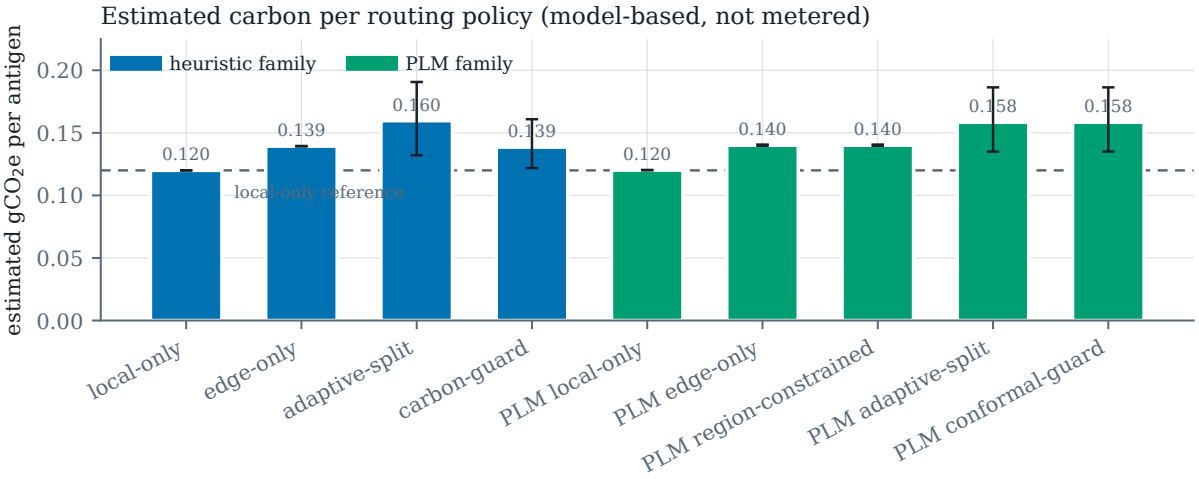


Figure 3: Modelled carbon per antigen by policy, with 95% intervals. These are outputs of the energy model described in Section 3.4, computed from fixed device power, an amortised embodied term and measured latency; they are not metered measurements. Uncertainty-triggered routing is the most carbon-intensive policy because a routed antigen is charged two device-requests.

Table 4: Provenance of each reported cost quantity. Only latency is measured; the energy and carbon figures are derived from fixed device power and a grid-intensity constant, and are therefore model outputs rather than measurements.

Quantity	Provenance
Latency	timed benchmark execution
Energy	estimated from fixed power and latency
Operational carbon	estimated from energy and grid intensity
Embodied carbon	amortized estimate from device metadata

That is the offloading energy threshold of Kumar and Lu (2010) written for this cost model, and our system falls on the wrong side of it because per-residue screening on short sequences makes the gating pass a large fraction of the work it gates. The same inversion has been reported for draft-and-verify decoding cascades (Dutta et al., 2026). What the measurements contribute is the location of the threshold for this task, not the existence of one.

The carbon-guard policy, which exists precisely to suppress unprofitable routing, does reduce carbon to 0.139 — essentially edge-only's figure — and cuts latency furthest, to 6.6 ms. It pays for this in quality: against adaptive-split its ΔMCC is -0.033 [-0.046, -0.022], an interval that excludes zero, and its MCC of 0.173 is statistically indistinguishable from doing no routing at all. On this evaluation the carbon guard does not find a useful middle point; it mostly turns routing off.

Within the PLM family the pattern is similar but compressed, and one component turns out to do nothing. The conformal-guard policy produces per-seed metrics identical to PLM adaptive-split on every metric we record, to the precision stored in the artifacts: the conformal criterion never changed a routing decision on this split. We report it as a null component rather than omitting it. The region-constrained post-processor, by contrast, is the one PLM addition that helps, improving on PLM local-only by ΔMCC +0.024 [+0.002, +0.045] at no latency cost relative to PLM edge-only.

4.2 Reproducing the previously reported fusion advantage

Before re-estimating the fusion effect we checked that any change could be attributed to the protocol rather than to our reimplementation. The original selection rule blends raw scores, scans

Table 5: The fusion effect under two selection protocols. The test-derived protocol chooses the mixing weight and the decision threshold on the same split it reports; the nested protocol chooses both inside the training folds of a leave-one-sequence-out loop. Intervals are 95% bootstrap intervals over per-antigen differences.

Split	Protocol	Weight	ΔROC-AUC	ΔMCC	Antigens improved
BP3C50ID external test	test-derived	0.90	+0.0018	+0.0115	–
	nested (LOSO)	0.90	-0.0005	-0.0329	7/15
		[0.55, 0.90]	[-0.0033, +0.0021]	[-0.0955, +0.0088]	
IEDB bounded subset	test-derived	0.95	+0.0012	+0.0012	–
	nested (LOSO)	0.95	+0.0008	+0.0014	25/47
		[0.85, 0.95]	[-0.0010, +0.0027]	[-0.0037, +0.0066]	

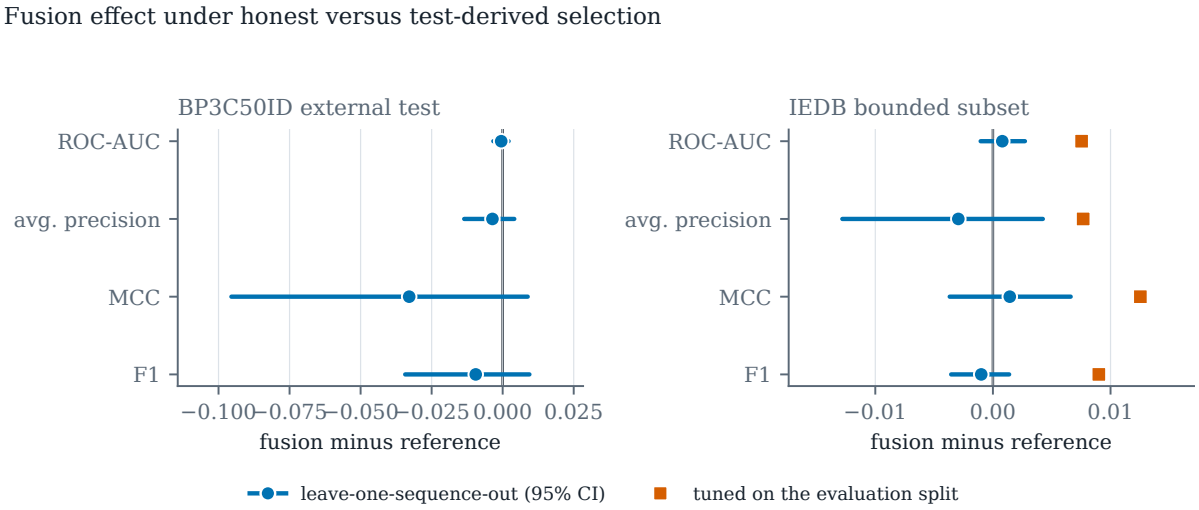


Figure 4: The fusion effect against the reference predictor under two selection protocols. Blue intervals are mean per-antigen differences with 95% bootstrap intervals under leave-one-sequence-out nested selection; orange squares are the corresponding estimates when the weight and threshold are chosen on the evaluation split. On both datasets the nested interval contains zero while the test-derived estimate sits above it.

the weight grid on the evaluation split, and sets the threshold to preserve the reference predictor's positive-call count on that split. Run through the loaders and metric code used throughout this paper, it reproduces the earlier analysis. All 44 entries of its stored weight-sensitivity grid agree to within 4.9e-07, inside the 1e-06 tolerance the check applies and consistent with the precision at which that grid was rounded. The procedure independently selects the same optimum, $w = 0.90$, and the same positive-call count of 1194. This comparison is executed by the replication script and its outcome recorded in the artifact (status: reproduced) rather than asserted here. Differences between the two protocols below are therefore attributable to how the configuration is chosen rather than to a difference in implementation.

4.3 The fusion advantage does not survive nested selection

Table 5 and Figure 4 give both protocols on both datasets.

On BP3C50ID, nested selection leaves a mean per-antigen difference against BepiPred-3.0 of -0.0005 ROC-AUC [-0.0033, +0.0021], with 7 of 15 antigens improving and a Wilcoxon p of 0.925. MCC behaves the same way (-0.0329 [-0.0955, +0.0088], $p = 0.561$).

The larger IEDB subset behaves the same way, with one important difference in what it tests. We have no BepiPred-3.0 predictions for those antigens, so that arm fuses the candidate with our own local-only predictor rather than with BepiPred-3.0. It is a second instance of the same

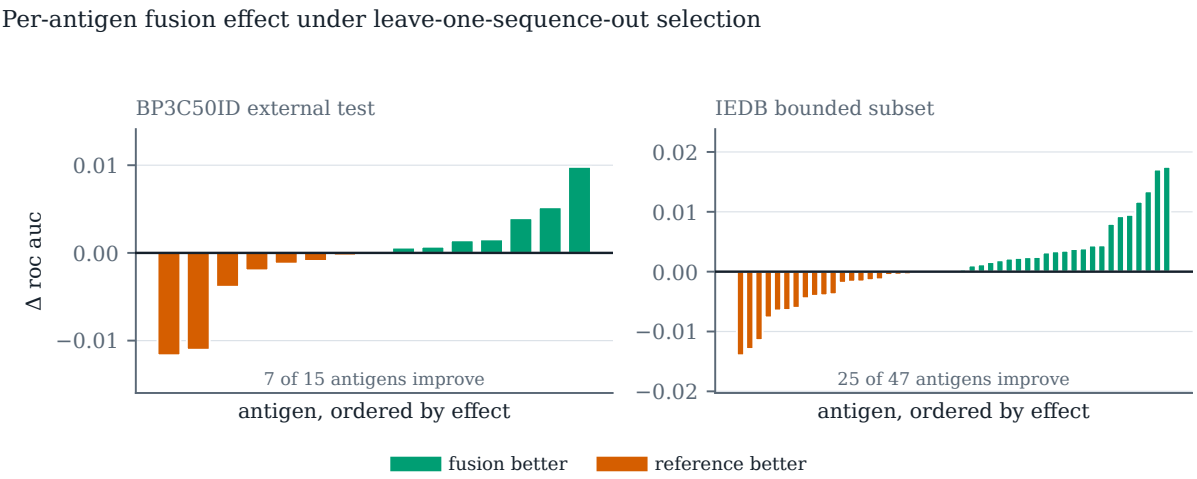


Figure 5: Per-antigen change in ROC-AUC from fusion under nested selection, ordered by effect. The dispersion, particularly on the IEDB subset, is much larger than the mean effect in either direction.

selection question, not a replication of the same comparison, and we do not read it as independent confirmation of the BepiPred-3.0 result. The mean per-antigen difference in ROC-AUC is +0.0008 [-0.0010, +0.0027], with 25 of 47 antigens improving ($p = 0.485$), and MCC gives +0.0014 [-0.0037, +0.0066]. Every interval on both datasets contains zero. Both the Wilcoxon test and the sign test discard antigens whose paired difference is exactly zero; on BP3C50ID this removes 1 antigens for ROC-AUC and 0 for MCC, and on the IEDB subset 0 and 8, so the effective sample size is smaller than the stated number of antigens by that count.

We are careful about what these nulls support. An interval containing zero is not evidence that the effect is zero, and at these sample sizes the paired design cannot resolve effects of the size at issue. With the observed per-antigen dispersion, the smallest difference BP3C50ID could detect with 80% power at a two-sided 5% level is 0.0822 in MCC and 0.0040 in ROC-AUC; on the IEDB subset the corresponding figures are 0.0073 and 0.0027. The MCC effect the test-derived protocol reports, +0.0115, is far below what the nested analysis could have detected. A non-significant nested result was close to guaranteed, so it carries little evidential weight against that specific effect.

What the design does identify is the behaviour of the estimator. Under the test-derived protocol the same BP3C50ID data yield +0.0115 MCC and +0.0018 ROC-AUC in favour of fusion; under nested selection the same quantity on the same data is -0.0329 and -0.0005. The point estimate does not shrink towards zero, it changes sign. That is a statement about how much the selection step moves the estimate, which this comparison does identify, rather than a statement about the underlying effect, which it does not. The defensible conclusion is that the test-derived estimate is biased by at least the distance between the two, not that fusion does not help.

Figure 5 shows the per-antigen differences behind these means. The dispersion is far larger than the mean in both directions. That is what a null effect over a small number of antigens looks like, and it is the reason we report the paired tests rather than a pooled number alone.

4.4 Which predictor the procedure prefers

Figure 6 plots the weight chosen in each training fold. On BP3C50ID the folds settle around a median of 0.90 (range 0.55 to 0.90); on the IEDB subset the median is 0.95 (range 0.85 to 0.95). Both sit close to the upper end of the grid, meaning the selection procedure chooses to give our predictor almost no influence over the blended score.

The selected weights are an independent route to the same conclusion. The nested evaluation

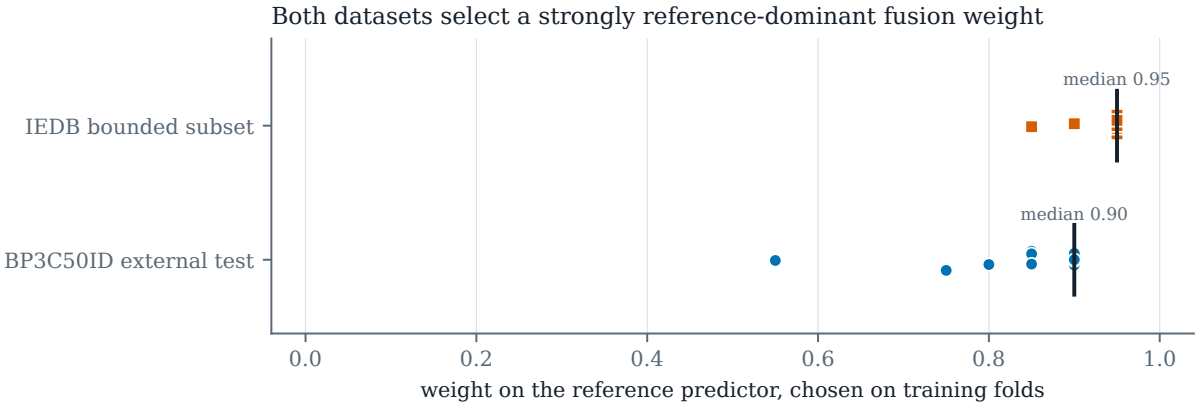


Figure 6: The fusion weight selected in each training fold, by dataset, with medians marked. Both datasets place almost all of the weight on the reference predictor, which is the same conclusion the null effect reaches by a different route.

says the blend cannot be distinguished from the reference predictor. The selected weights say why: the procedure, given a free choice, declines to use the second predictor. A fusion whose fitted weight approaches one is not a combination of two models but the stronger model with extra machinery attached.

5 Discussion

5.1 Routing is a disclosure and latency mechanism, not a green one

The case for local–edge routing in a screening pipeline is sound, but it is a different case from the one usually made, and narrower than we expected to be able to make. On our evaluation the mechanism buys two things a laboratory can verify directly, and both survive antigen-level resampling. Screening finishes in roughly a third of the time always-remote execution takes. Most antigens are never transmitted. The second gain is not obtainable by buying a faster remote service.

What it does not buy, on evidence of this size, is a demonstrated quality position. Against always-remote execution the quality difference is not resolved; against always-local execution only one metric of four separates them. We had originally read the same comparisons as decisive, on intervals computed over seeds, and Section 4.1 explains why those were the wrong intervals. A laboratory deciding whether to adopt routing should therefore treat the latency and disclosure gains as established here and the quality question as open. That is a less satisfying recommendation than the one we set out to make. It is also a more defensible one.

What the mechanism does not buy is lower energy, and Equation 2 says precisely when it would. Offloading saves energy only above a ratio of work offloaded to overhead incurred (Kumar and Lu, 2010). Our contribution is to locate that boundary for this task, at a routed fraction of 0.139, and to observe that a policy tuned for predictive quality routes about twice as often as that. The green argument would recover for longer antigens or a heavier remote tier, where the ratio shifts. Practitioners can evaluate Equation 2 for their own pair rather than assuming either direction.

The practical recommendation, then, is to adopt routing when sequence disclosure or turnaround is the binding constraint, and not to adopt it on environmental grounds without measuring the ratio for the models actually in use.

5.2 What the fusion result means, and what it does not

Our result is that a convex blend of CARBON-BCE-R with BepiPred-3.0 shows no advantage over BepiPred-3.0 alone, on either of two datasets, once the blend is chosen without reference to the evaluation labels. The scope of that claim matters.

It is not a claim that score fusion is useless in epitope prediction. It is a claim about one pair of predictors on two splits. One member of that pair is considerably weaker than the other, and the folds consistently chose to down-weight the weaker member. A fusion of two comparably strong and genuinely complementary predictors might well behave differently. Nothing here tests that.

It is also not a claim that the earlier result was dishonestly produced. The protocol that generated it is easy to arrive at by accident, because a weight sweep serves two different purposes that look the same on the page. Presented as a sensitivity analysis, it answers "how fragile is this result to the weight?" Used to pick the reported row, it answers "what is the best this combination can look like?" The second question's answer is then reported as though it were the first's. The diagnostic is procedural rather than statistical: ask whether the configuration being reported was selected using the labels it is scored against.

What the result does establish is that in this corner of the literature the difference between the two protocols is large enough to manufacture a positive finding out of a null one, on data where the optimistic protocol produced a plausible-looking effect with a bootstrap interval attached. Reporting an interval does not protect against selection bias when the selection happened before the interval was computed. Given how inexpensive the nested alternative is — one extra loop over sequences, no new data, no new model — we think it should be the default whenever a combination parameter is tuned on a dataset of this size.

5.3 Relation to prior work

Our benchmark is built on the BP3C50ID split distributed with BepiPred-3.0 (Clifford et al., 2022). The local predictor descends from the feature-based and deep approaches surveyed by Cia et al. (2023) and exemplified in this journal by Attique et al. (2023). We do not advance the state of the art in predictive accuracy and do not claim to. Standalone CARBON-BCE-R remains below BepiPred-3.0 on BP3C50ID, and structure-aware methods such as DiscoTope-3.0 (Høie et al., 2024) occupy ground our sequence-only setting does not reach.

Our evaluation is small beside purpose-built benchmarks for this area such as AsEP (Liu et al., 2024), and we call it a measurement harness rather than a benchmark for that reason. Within the scope of inference routing, however, we are not aware of prior work that routes between a local and a remote conformational epitope predictor and measures the consequences. Federated approaches to protein modelling address a different problem. They distribute training rather than place inference.

The systems contribution sits alongside the split-computing literature (Matsubara et al., 2022), from which the routing mechanism is borrowed rather than invented. What we add is the measurement of that mechanism against the objectives a wet-lab screening workflow actually has. That includes the counting measure of raw-sequence disclosure, which accuracy-oriented formulations do not express, and the observation that the energy argument inverts at this scale.

The methodological contribution is an instance of the problem Kapoor and Narayanan (2023) document across applied machine-learning fields, reported from the inside: we are correcting our own earlier result rather than auditing someone else's. We think instances of this kind are useful precisely because the published record contains few of them.

6 Limitations

Several constraints bound what can be concluded from this study.

Carbon is modelled, never metered. Every energy and carbon figure in this paper is computed from a fixed device power constant, a measured latency and a grid-intensity constant (Section 3.4, Table 4). No wall-power measurement was taken at any point. The carbon comparisons between policies are therefore comparisons of modelled quantities that are proportional to measured latency; they support statements about which policy does more work, and they do not support any statement about emissions of a deployed system. A metered replication on instrumented hardware is the obvious next step and is not something this study attempted.

The carbon figures rest on two utilisation constants. The energy model reproduces, but only once the utilisation fractions are supplied by hand. Across all 47 antigens of the second dataset, recomputing each per-antigen total from the recorded device configuration, the measured latencies and the utilisation fractions of Section 3.4 matches the stored values to within 0e+00 gCO_2e. The fractions are not in the recorded configuration, so the stored totals cannot be rebuilt from that configuration alone, and the released decomposition artifact says so. The fractions themselves, however, are literals in the benchmark code rather than entries in any configuration file, and they dominate the result: the embodied term is almost the whole of each per-request total.

That matters for how far the carbon conclusion travels. Of 5 alternative settings we evaluated, the ordering survives in all but 1. A device pair with the local tier at 0.10 utilisation and the remote tier at 0.90 would put the break-even routed fraction at 0.690, above the rate our policy actually routes at. Routing would then be the greener option. Our finding therefore holds for the hardware pair modelled here and for most reasonable variations on it. It remains a statement about a cost model whose dominant parameters deserve to be measured rather than assumed. A metered replication remains the right next step.

The remote tier is co-located, not remote. The latency figures attribute a transport cost to routed antigens, but the benchmark executes both tiers on one machine, and the two-node transport evidence in the artifact is an HTTP probe rather than a deployed remote model server. Real deployments would add network variance, queueing and failure modes that this harness does not model. For latency and for transmitted volume these would, if anything, favour routing rather than count against it, since always-remote execution pays the network on every antigen and routing pays it on roughly a quarter. What a real deployment would add against routing is of a different kind: a dependency on the remote tier for exactly the antigens the local model found hardest, and the failure and retry behaviour that comes with it. We measure none of that.

The primary benchmark is small, and that bounds nearly everything. BP3C50ID's external split has 15 antigens. Resampling on 15 units gives intervals wide enough that most quality comparisons in this paper are not resolved, and every such comparison should be read as "no effect detectable at this sample size" rather than as "no effect". This applies to the routing results as much as to the fusion analysis: the interval on the adaptive-split versus always-remote MCC difference spans roughly ± 0.05, which would not exclude a quality loss a practitioner would care about. For the fusion analysis the IEDB subset at 47 antigens carries more weight on ROC-AUC and average precision, where it has no tied folds. On the thresholded metrics its informative-fold count is closer to BP3C50ID's and we do not claim it is better powered there.

The language-model tier is the smallest available checkpoint. The PLM results use `esm2_t6_8M_UR50D`, chosen for the hardware at hand. Conclusions about the PLM family are conclusions about an 8M-parameter frozen encoder and should not be read as characterising what language-model features can contribute to this task.

One routing signal, one validation design. Routing decisions come from mean per-residue local uncertainty, selected on a single grouped validation split per seed. Repeated grouped resampling of the validation split would give a better picture of how stable the selected thresholds are; we vary the seed, which moves the split, but we do not separate the two sources of variation.

The IEDB subset's sampling rule is not recorded. Its source and identity filter are established (Section 3.1), but how the 47 test and 205 training antigens were drawn from the 3560 available is not captured in any script we could find. The selection is therefore reproducible in its inputs but not in its sampling. We state this rather than reconstruct a rule after the fact, since a plausible-looking rule invented now would be indistinguishable in the write-up from one that was actually followed, which is the failure this paper is about. What we can do is pin down the set: the released manifest lists every test antigen by accession, so the evaluation subset is identified exactly even though the draw that produced it is not. The consequence for the reader is that the IEDB arm should be treated as a second dataset of opportunity rather than as a designed replication, and we do not rest the paper's conclusions on it.

Declarations

Data and code availability. The benchmark harness, the analysis modules that produce every table and figure in this paper, and the raw per-seed result artifacts are available at `https://github.com/Anandharajan/edgeconbce`. A single command (`regenerate.sh`) rebuilds the generated-number file, all tables, all figures and this manuscript from the stored results. The repository is archived on Zenodo at acceptance and the DOI will replace this URL in the published version.

The BP3C50ID training and external test splits, and the `IEDBSeqsNotSharedAt20ID.fasta` collection from which the second evaluation set is drawn, are redistributed by the Technical University of Denmark alongside BepiPred-3.0. We do not republish them, and the repository contains no antigen sequences from either source. The BepiPred-3.0 predictions used as the reference were obtained from the official service.

The rule by which the 47 test and 205 training antigens were drawn from that collection was not recorded when the subset was made, and we have not been able to recover it. We therefore release the identity of the evaluation subset instead (`data/iedb_subset_manifest.csv`), which lists every test antigen by accession, so that the set itself can be reconstructed from the source collection even though the draw cannot be repeated. The 205 training antigens are counted in the released artifacts but not identified, and we do not reconstruct them after the fact.

Ethics. This study uses only published protein sequence data. No human participants, human tissue, animal subjects or personally identifying information were involved, and no ethics approval was therefore required.

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Declaration of generative AI in the writing process. During the preparation of this work the author used generative AI tools to assist with the work. The author reviewed and edited all content, verified every result reported here, and takes full responsibility for the content of the published article.

CRediT author statement. Anandharajan TRV: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review and editing, Visualization, Supervision, Project administration.

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